

SHOULD-COST ANALYSIS REPORT

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Bottom-Up Manufacturing Cost Model · Fully Traceable Rate Data

Flange with Central Boss and 6-Hole Bolt Pattern

Commodity: MACHINING · Currency: CNY · FX Rate: 9.0498 to GBP · Operations: 5 · Region: CN

Total Should-Cost	Material	Process	Labour	Margin
¥177.31	16.6%	46.8%	18.1%	7.4%

Model Confidence: **Low** · 5 traced operations · 12 data points auditable

Engine Warnings (1)

rawMaterial.materialId: Material rate confidence: Medium

Geometry Provenance — MEASURED (OCCT B-rep kernel)

Volume, weight and every feature were measured from the CAD solid by the Open CASCADE kernel. Measured volume 63.9 cm³; mass for the selected material family 0.173 kg.

Material cost and geometry-derived machining are grounded in the actual solid — not an estimate.

Key Assumptions

Annual volume: 200,000 · Region: CN · Commodity: machining

Alloy / material: 6061-T6 · Net weight: 0.502 kg (measured)

Material utilisation: 50% · Overhead: 12% of factory base · Margin: 8% of subtotal · Operations: 5

§1 — 8-Bucket Cost Breakdown

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
1. Raw Material	¥29.48	16.6%	
2. Process (Machine)	¥83.02	46.8%	
3. Direct Labour	¥32.11	18.1%	
4. Tooling (amortised)	¥0.68	0.4%	
5. Packaging	¥0.63	0.4%	

Cost Bucket	Amount (CNY)	% of Total	Cost Mix Bar
6. Logistics	¥0.81	0.5%	
Factory Cost	¥146.74	82.8%	
7. Overhead (SG&A)	¥17.43	9.8%	
Subtotal	¥164.17	92.6%	
8. Supplier Margin	¥13.13	7.4%	
TOTAL SHOULD-COST	¥177.31	100.0%	

§2 — Commercial Parameters			
Overhead Rate	12.0%	Supplier Margin	8.0%
Packaging / part	¥0.63	Logistics / part	¥0.81
Total Tooling Cost	¥135746.61	Amortisation Volume	200,000 parts

Geometric DFM / DFA - measured from the CAD model

1 issue(s) across 1 instance(s), from 6 measured feature(s) and 3 rule(s). Every issue names the B-rep faces that produced it and the published source of its threshold. Priced findings total ¥8.80/part; issues shown without a figure are quality or yield risks with no modelled cost path — see each entry.

[MAJOR] Hole deeper than standard drill reach
¥8.80/part

Worst: # 6.0 mm hole is 72.0 mm deep — 12.0:1 depth-to-diameter.
measured IdRatio 12:1 threshold > 5:1
Faces: 4, 5, 6, 7, 8, 9
Fix: Beyond ~10:1 this is a gun-drilling or trepanning operation. Open the diameter, drill from both ends, or accept a specialist operation and its setup.
Source: Machinery's Handbook — drilling: depth-to-diameter limits and standard drill sizes
Standard jobber drills reach ~5xD; parabolic / coolant-through tooling reaches ~10xD; beyond that gun drilling. Reported at the point the process and the cost step.

DFA - handling & insertion (Boothroyd method, geometric half)

Handling 1.38s + insertion 1.5s = 2.88s, 0.96x the 3s ideal part.
+0.25s Partial symmetry — one orientation decision remains (bounding box 80×80×20 mm)
Theoretical minimum part count, and therefore the Boothroyd design-efficiency index, requires the full assembly — this analysis covers one part, so no efficiency ratio is given.
Nesting and tangling risk is not assessed: it depends on how parts are presented in bulk, which the geometry alone does not determine.
Rotational symmetry is approximated from the bounding box, not from an inertia-tensor shape comparison.

What was NOT checked

No material family was confirmed, so any check whose threshold depends on the alloy or resin did not run.
Tolerance and surface-finish callouts were not checked — they are on the drawing, not in the solid, and they drive whether a feature is milled, ground or honed.
Tool access and fixturing were not simulated: a feature can be geometrically fine and still unreachable in the chosen setup.
Thread specifications were not verified against standard tap sizes.
Workholding and part distortion under clamping were not assessed.

§3 — Material Detail

Parameter	Value	Unit	Notes
Material ID	mat-al6061	ID	
Grade / Specification	6061-T6		LME + UK processor premium, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.970 (exchangeMetal)
Region	China		
Net Finished Weight	0.5020 kg	kg	Weight in finished part
Gross Weight (stock)	1.0040 kg	kg	Net ÷ utilisation ratio
Scrap / Runner Weight	0.5020 kg	kg	Gross - Net
Material Utilisation	50.0%	%	Benchmark: machining 60–75 % (stock removal)
Material Price	¥31.78	CNY /kg	LME + UK processor premium, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.970 (exchangeMetal)
Scrap Recovery Price	¥4.83	CNY /kg	
Gross Material Cost	¥31.90	CNY	Gross × price/kg
Scrap Credit	-¥2.42	CNY	Scrap × recovery price
NET RAW MATERIAL COST	¥29.48	CNY	Gross - scrap credit + consumables
Data Confidence	Medium		
Effective Date	2026-07		

§4 — Operations Detail

4A Machine Operations

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
Setup (amortised)	CNC Lathe (2-axis)	¥185.20	0.60	1	100.0 %	¥1.85
Turning — +X (8 faces)	CNC Lathe (2-axis)	¥185.20	10.21	1	85.0%	¥37.06
Turning — +Z (3 faces)	CNC Lathe (2-axis)	¥185.20	3.81	1	85.0%	¥13.85
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	CNC Drilling Centre	¥136.82	7.22	1	85.0%	¥19.36
Drilling — 6 holes	CNC Lathe	¥185.20	3.00	1	85.0%	¥10.89

Operation	Machine Class	Rate / hr	Cycle (min)	Parts / Cycle	OEE %	Machine Cost
(6×Ø6.0×12) [geometry-measured]	(2-axis)					
TOTAL						¥83.02

4B Labour Detail

Operation	Labour Grade	Rate / hr	Man ning	Lab. min	Eff %	Labour Cost	Op Total
Setup (amortised)	Skilled Machinist	¥71.49	1	0.60	100.0%	¥0.71	¥2.57
Turning — +X (8 faces)	Skilled Machinist	¥71.49	1	10.21	92.0%	¥13.22	¥50.28
Turning — +Z (3 faces)	Skilled Machinist	¥71.49	1	3.81	92.0%	¥4.94	¥18.79
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	Skilled Machinist	¥71.49	1	7.22	92.0%	¥9.35	¥28.71
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]	Skilled Machinist	¥71.49	1	3.00	92.0%	¥3.89	¥14.78
TOTAL						¥32.11	¥115.13

4C Machined Features — Geometry Audit

Feature	Ø×depth / area	Qty	Operation	Min	Cost	Costed ?
Boss	Ø30.0 × 8	1	Turning (external Ø)	0.46	¥1.95	Yes
Boss	Ø80.0 × 12	1	Turning (external Ø)	0.54	¥2.29	Yes
Hole / bore	Ø6.0 × 12 blind	6	Drilling (blind)	1.68	¥7.12	Yes
Face	4857 mm²	1	Face milling (facing)	0.81	¥3.42	Yes
Face	4150 mm²	1	Face milling (facing)	0.72	¥3.05	Yes
Face	707 mm²	1	Face milling (facing)	0.29	¥1.22	Yes
TOTAL costed		11		4.49	¥19.05	Yes

§5 — Machine Rate Buildup

CNC Lathe (2-axis) [mach-lathe-cnc] · Confidence: Low · UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥199095.02	¥62.22	4,000 hr/yr available
Maintenance	¥109502.26	¥34.22	
Energy	¥55085.58	¥17.21	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥109502.26	¥34.22	
Finance Cost	¥79638.01	¥24.89	
TOTAL — CNC Lathe (2-axis)	¥592642.14	¥185.20	Effective hours: 3200/yr

CNC Drilling Centre [mach-drill] · Confidence: Low · UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 | Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh

Cost Component	Annual Cost (CNY)	Rate/hr @80% util	Notes
Depreciation	¥139366.52	¥43.55	4,000 hr/yr available
Maintenance	¥59728.51	¥18.67	
Energy	¥49577.02	¥15.49	
Floor Space	¥39819.00	¥12.44	
Indirect Support	¥79638.01	¥24.89	
Finance Cost	¥69683.26	¥21.78	
TOTAL — CNC Drilling Centre	¥437812.32	¥136.82	Effective hours: 3200/yr

How to read the machine rate

Rate/hr = (annual depreciation + maintenance + energy + floor + indirect + finance) ÷ effective hours, where effective hours = available hours × utilisation.

Effective hours encode the shift pattern: a lower hours-base raises £/hr and a higher one lowers it. A challenge from a supplier will target this divisor and whether depreciation is machine-only or full-cell (machine + dosing furnace + trim + robots/extraction) — state your basis when defending the number.

§6 — Rate Traceability

Field	Value	Unit	Source / Reference	Rate ID	Conf
material.pricePerKg	3.5114	£/kg	LME + UK processor premium, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.970 (exchangeMetal)	mat-al6061	Medium

Field	Value	Unit	Source / Reference	Rate ID	Conf.
material.scrapRecoveryPricePerKg	0.5335	£/kg	LME + UK processor premium, Jun 2026. Index-anchored 2026-07 refresh. Regional adj. ×0.970 (exchangeMetal)	mat-al6061	Medium
Setup (amortised).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Setup (amortised).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Turning — +X (8 faces).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Turning — +X (8 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Turning — +Z (3 faces).machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Turning — +Z (3 faces).labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].machineRatePerHr	15.1182	£/hr	UK Tier-2 CNC drilling/tapping centre. Target £30/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-drill	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].machineRatePerHr	20.4647	£/hr	UK Tier-2 CNC turning centre (Doosan/Hyundai class). Target £40/hr. Jun 2026 Regional: capex/overhead ×0.55, energy re-tariffed @£0.07/kWh	mach-lathe-cnc	Low
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured].labourRatePerHr	7.9000	£/hr	Regional benchmark China — 2026 Q2	lab-uk-skilled	Low

§7 — Should-Cost with Uncertainty

Monte-Carlo · Low confidence

Estimate (± band)	P10 · optimistic	P50 · median	P90 · conservative	Band	CV
¥177.31 ± 18%	¥146.33	¥175.48	¥210.23	wide	14.2%

Range reflects how well each cost driver is known (High / Medium / Low confidence). Tighten it by attaching CAD, a BOM, or logging an actual quote.

Why confidence is Low

Main band drivers: 10 rate(s) at Low confidence; tooling amortisation carries the widest per-bucket spread.

What's excluded from this unit cost

One-time NRE — tooling / die, fixtures and CNC programming — is amortised separately, not in this unit cost.

Import duty and international freight (regional table is Ex-Works).

Formal embodied-carbon reporting (figures are indicative cradle-to-gate — replace with supplier EPDs).

§8 — Sensitivity Analysis

± 10% driver swing · ranked by ¥ impact

Cost Driver	Base	-10% cost	+10% cost	Range ¥
Setup (amortised): Machine Rate	¥185.20/hr	-4.3%	+4.3%	¥15.40
Turning — +X (8 faces): Machine Rate	¥185.20/hr	-4.3%	+4.3%	¥15.40
Turning — +Z (3 faces): Machine Rate	¥185.20/hr	-4.3%	+4.3%	¥15.40
Drilling — 6 holes (6×Ø6.0×12) [geometry-measured]: Machine Rate	¥185.20/hr	-4.3%	+4.3%	¥15.40
Turning — +X (8 faces): Cycle Time	0.17hr	-3.4%	+3.4%	¥12.16
Setup (amortised): Labour Rate	¥71.49/hr	-2.2%	+2.2%	¥7.77
Turning — +X (8 faces): Labour Rate	¥71.49/hr	-2.2%	+2.2%	¥7.77
Turning — +Z (3 faces): Labour Rate	¥71.49/hr	-2.2%	+2.2%	¥7.77

§9 — Regional Cost Comparison

Ex-Works · per-region should-cost · vs China

Region	Material	Process	Labour	Tooling	Overhead	Ex-Works	Logistics	Total	vs China
United Kingdom (GBP)	¥33.50	¥150.95	¥115.59	¥1.23	¥23.25	¥324.52	¥0.56	¥339.12	+91%
Germany (EUR)	¥34.51	¥158.50	¥186.81	¥1.30	¥32.35	¥413.46	¥0.65	¥431.78	+144%
France (EUR)	¥34.17	¥138.88	¥134.27	¥1.14	¥24.99	¥333.44	¥0.65	¥348.54	+97%
Spain (EUR)	¥33.50	¥132.84	¥84.65	¥1.09	¥18.48	¥270.55	¥0.65	¥283.08	+60%
Poland (PLN)	¥32.50	¥108.69	¥52.54	¥0.89	¥12.76	¥207.37	¥0.67	¥217.28	+23%
Turkey (TRY)	¥30.15	¥90.57	¥29.19	¥0.74	¥9.07	¥159.72	¥0.73	¥167.67	-5%
China (CNY)	¥29.48	¥83.02	¥32.11	¥0.68	¥17.43	¥162.73	¥0.81	¥177.31	Base
India (INR)	¥30.15	¥78.50	¥20.43	¥0.64	¥7.21	¥136.93	¥0.84	¥143.93	-19%
Mexico (MXN)	¥31.83	¥90.57	¥33.86	¥0.74	¥9.45	¥166.45	¥0.76	¥174.65	-2%
United States (USD)	¥33.50	¥128.31	¥151.78	¥1.05	¥23.06	¥337.71	¥0.70	¥352.94	+99%

Per-region should-cost using regional labour, energy and rate benchmarks. Tooling held fixed. Ex-Works — excludes import duty + international freight. Indicative — confirm with RFQ.

§10 — Embodied Carbon

5.82 kgCO2e · 11.59 /kg · cradle-to-gate

Material	4.96 kgCO2e	Material factor	8.60 kg/kg (Aluminium primary mix)
Process	0.77 kgCO2e	Grid intensity	0.550 kg/kWh · 1.41 kWh
Logistics	0.08 kgCO2e	Total	5.82 kgCO2e

Indicative cradle-to-gate factors — replace with supplier EPDs for formal reporting.

[WARNING] Machining / process cost above benchmark

Finding	Process cost at 46.8% of total is above the benchmark range of 18–35%. Cycle time or machine rate is elevated.
Impact	High · up to 6% potential saving
Benchmark	Process %: yours 46.8% vs industry 18–35%
Action 1	Reduce cycle time via feeds/speeds optimisation and toolpath strategy review
Action 2	Combine multiple operations into one setup to eliminate re-fixturing time

[WARNING] Low material utilisation — high scrap rate

Finding	Material utilisation at 50% is below the machining benchmark of 72%. You are paying for 100% more raw material than ends up in the part.
Impact	Medium · up to 4% potential saving
Action 1	Redesign blank/preform geometry to minimise offcuts
Action 2	Optimise nesting/layout for sheet or profile stock

[INFO] Material cost is below benchmark range

Finding	Material at 16.6% is below the typical 30–55% range. Conversion cost is the dominant driver.
Impact	Low
Benchmark	Material %: yours 16.6% vs industry 30–55%
Action 1	Focus optimisation efforts on process and labour efficiency

§12 — Cost-Reduction Opportunities (ranked by saving)

7 opportunities · combined ¥31.92/part (~18%)

Ranked on money per part against this costing. The combined figure is the root-sum-square of the top three, capped at 40% — overlapping actions do not save twice, so the column is deliberately not summed.

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
1	Material	Improve Material Utilisation via Near-Net-Shape	Current utilisation at 50%. Near-net-shape process or improved nesting targets 80–90%.	¥26.60	15.0 %	Medium Term	Medium
2	Material	Scrap Revenue Recovery at Index Prices	50% of the bought material leaves as chips, offcuts or skeleton. That scrap has an index price — the credit belongs in the piece price, not in the supplier's margin.	¥7.09	4.0%	Quick Win	Low

#	Category	Action	Why it is on the list	Save/p art	%	Timeframe	Risk
3	Process & Automation	Consider near-net-shape pre-form (casting/forging)	Process cost dominates (>40%) — Process cost at 46.8% of total. Machining conversion cost is unusually high.	¥14.18	8.0%	Medium Term	Medium
4	Process & Automation	Unmanned / Lights-Out Running	Labour is 18.1% of cost on a machine-paced part with healthy OEE (88%) — the classic profile for bar-feed or pallet-pool unattended running.	¥9.63	5.4%	Medium Term	Medium
5	Process & Automation	Multi-Machine Manning	Operations are manned at up to 1.0 operators per machine while the machine paces the cycle. One operator tending two machines is the textbook correction.	¥8.03	4.5%	Quick Win	Low
6	Design & Geometry	Part-Count / Feature Integration Review (Boothroyd)	5 operations suggest features or joined parts that may not need to exist separately. The Boothroyd minimum-part-count test asks of each: does it move, must it be a different material, must it be separate for assembly?	¥10.64	6.0%	Long Term	Medium
7	Design & Geometry	Tolerance and Surface-Finish Relaxation	Machining is 46.8% of the part. The tightest tolerance and finish callouts set the machine, the cycle and the inspection burden — reviewing them is the cheapest cost lever that exists.	¥7.09	4.0%	Quick Win	Low

§13 — Opportunity by Category

3 categories with an actionable lever

Category	Opportunities	Best single action	Best save/part	Category total (indicative)
Material	2	Improve Material Utilisation via Near-Net-Shape	¥26.60	¥33.69
Process & Automation	3	Consider near-net-shape pre-form (casting/forging)	¥14.18	¥31.84
Design & Geometry	2	Part-Count / Feature Integration Review (Boothroyd)	¥10.64	¥17.73

§15 — Implementation Roadmap

Quick Wins — Implement Immediately

- Multi-Machine Manning
- Scrap Revenue Recovery at Index Prices
- Tolerance and Surface-Finish Relaxation

Long-Term Changes — Strategic Investment

- Part-Count / Feature Integration Review (Boothroyd)